

TAM analysis of interactive video acceptance and learning retention in higher education

Hamed Hosseini Zarrabi^a, Morteza Rezaei-Zadeh^{b,c,*}, Reza Nabovati^d

^a Faculty of Entrepreneurship, University of Tehran, Tehran, Iran

^b Shahid Beheshti University, Velenjak, Tehran, Iran

^c University of Leicester, Leicester, United Kingdom

^d Linköping University, 581 83, Linköping, Sweden

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ABSTRACT

This study addresses a critical gap in the educational technology literature by empirically comparing the effects of linear and interactive instructional videos on students' technology acceptance and learning retention in an informal learning environment. Employing a mixed-methods design, a within-subjects experiment was conducted with 55 university students who engaged with both video formats. Quantitative data, collected via a Technology Acceptance Model (TAM) questionnaire and a multiple-choice retention test, were analyzed using a paired *t*-test. The results show that students' technology acceptance ($t(54) = -10.31, p < .001$) and learning retention (mean score of 5.60 vs. 4.56) were significantly higher for the interactive video format. The qualitative phase, which involved twelve semi-structured interviews, identified seven key mechanisms that explain these positive outcomes, including enhanced perceived usefulness, improved ease of use, increased engagement, and support for personalized learning. These findings contribute to the field by providing a clear rationale for the pedagogical superiority of interactive videos and offering a practical framework for educators and designers to create more effective and engaging digital learning materials.

1. Introduction

The use of video for educational purposes dates back over eighty years [1]. With the rise of e-learning during the Covid-19 pandemic, video adoption accelerated due to its distinctive features and effectiveness [2–4]. Videos create an audiovisual learning environment that helps learners acquire skills and internalize knowledge [5,6]. By engaging multiple senses, this format can enhance memory and learning [7]. Some studies also evaluated students' adoption and behavioural intention to use such video lessons [8].

Despite these benefits, the use of video for educational purposes is not without its critics [9]. For instance, instructional videos often fail to meet educational needs because they are not interactive or manageable [7]. Played linearly, videos only allow for simple forward and backward navigation, offering no way to ask questions or engage with the content at key moments [10]. Many studies have identified a lack of interactivity as a key limitation of instructional videos [2], recommending that linear videos be upgraded to interactive formats to improve learning effectiveness [11,12] and increase learner satisfaction [13,14]. In light of the

critical importance of interactivity in educational content, an analysis of 190 existing digital products has revealed significant deficiencies pertaining to interactivity and learner autonomy. These findings have subsequently been used to establish a set of guidelines for designers of future interactive and self-directed digital educational resources intended for distance learning [15] while learners mostly preferred to interact more with the content in their distance courses [16].

Interactive videos are strongly supported by two key theories: constructivist theory and cognitive formation processing theory. Constructivism posits that learners play an active role in their own education through exploration and analysis [17]. Cognitive theory highlights how learners process and retain information. An interactive medium is particularly helpful for learners who, by nature, have limited attention spans [18,19]. Modern interactive videos offer functions like fast-forward, rewind, speed-up, and slow-down, allowing users to navigate freely and review specific parts as needed [20,21]. They can also embed various forms of information, such as text, images, and links, which makes them more appealing and provides a better learning environment [20,22,23].

* Corresponding author.

E-mail addresses: hamed.zarrabi@gmail.com (H. Hosseini Zarrabi), mrz5@leicester.ac.uk (M. Rezaei-Zadeh), R.nabovati@gmail.com (R. Nabovati).

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Scientific interest in interactive videos has grown significantly in recent decades. A Scopus search (on August 6, 2025) for “interactive video” in all fields returned 16,552 documents published between 1973 and 2025. As illustrated in Fig. 1 below, studies on this topic have increased over time, with approximately 60% of them concentrated in the fields of engineering, computer science, and medicine.

Recent studies have explored the effects of interactive videos on various aspects of student learning. While some research shows positive effects on student mood [24] and decision-making abilities (Taslibevaz et al., 2017), other studies have found no significant effect on technical proficiency [25]. Interactive videos have also been shown to support self-paced learning (Palajeergirou and Papadopoulou, 2019), reflective thinking [26], and active engagement [27,10]. Additional benefits include the facilitation of immediate feedback [28] and increased enjoyment [29].

Interactive video lectures integrate various activities between video segments [30], such as hotspots, content-related questions, and questionnaires. According to Sauli et al. [23], an interactive video is a “non-linear video that presents both classical (e.g., play, pause, stop and rewind/forward buttons) and more complex (e.g., table of contents and index) functions to control the navigation of the video stream,” often with embedded hyperlinks. This format facilitates the understanding of complex topics by breaking them into manageable segments, while also reducing the cognitive load on learners by presenting information visually.

Interactive videos are characterised by features such as non-predefined learning paths, free navigation, clickable elements, and the ability to embed external information [31]. Interactions can also be categorised based on their pedagogical purpose, such as rhetorical or inductive questions [32]. The most common design patterns for interactive video include watching specific parts, marking sections, embedding questions, and linking to internal or external content [33].

While a significant body of theoretical research exists on instructional videos, there are few experimental or quasi-experimental studies comparing different video formats [34]. In particular, empirical evidence is lacking on how linear and interactive videos affect learners' technology acceptance and learning retention, especially in informal learning environments. In the literature, informal learning environments are considered important because they extend beyond the structure of formal classrooms and provide self-directed, flexible, and

interest-driven learning opportunities tailored to learners' needs [35, 36]. This is particularly significant when comparing instructional approaches such as linear and interactive videos, as informal contexts better reflect authentic learning behaviors outside institutional settings [37,38] and therefore yield findings that are more representative of real-world user experiences [39].

Addressing this research gap is important because applying the Technology Acceptance Model (TAM) to interactive educational videos, particularly in informal contexts that better reflect authentic learning behaviors outside institutional settings, can strengthen the model's validity and generalisability within digital learning environments based on emerging educational technologies. Furthermore, identifying the determinants of educational video acceptance can assist e-learning instructional designers in developing more effective and engaging interactive content. To address this need, the present study adopts a qualitative approach aimed at deeply exploring the underlying mechanisms through which interactive videos influence students' technology acceptance and learning retention. This study aims to empirically examine how interactive versus linear video formats affect students' technology acceptance and learning retention, with a specific focus on uncovering the mechanisms that explain these effects.

The remainder of this paper is structured as follows: we first establish the theoretical foundation by reviewing key literature on the Technology Acceptance Model (TAM) and learning retention of video as a learning medium. Next, we detail our experimental methodology and present the data findings. The results are then analyzed through a pedagogical lens, followed by a discussion of their practical implications and study limitations. Finally, we conclude with recommendations for future research.

1.1. Hypotheses development

1.1.1. Technology acceptance

The technology acceptance model (TAM), developed by Fred Davis, is a well-known framework for explaining why people use new technology [40]. It posits that a user's perceived usefulness and ease of use determine their level of acceptance [41]. TAM has proven to be a reliable tool for evaluating technology acceptance in the IT industry [42].

Originally proposed in 1986, TAM identifies attitude and behavioral intention as key dimensions leading to technology acceptance. The

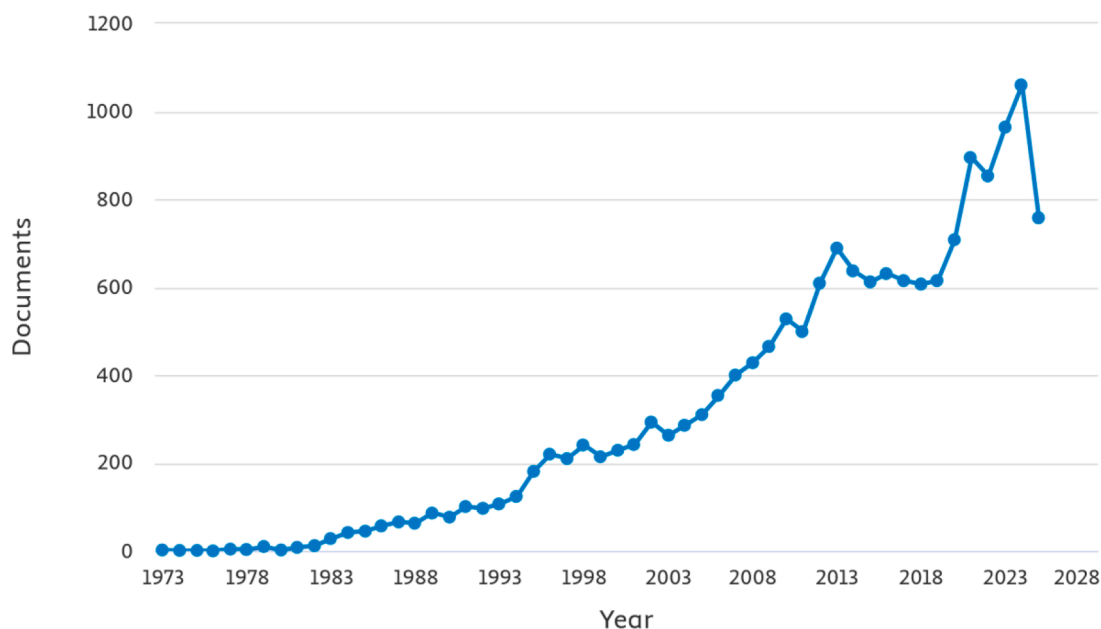


Fig. 1. number of studies conducted on interactive videos between 1973 and 2025.

model suggests that the easier a technology is to use, the more useful it is perceived to be. Therefore, a user's attitude and perceived usefulness are strong predictors of system adoption [43].

Nagy [40] extended the TAM to study student video usage and satisfaction, finding that video usage directly impacts perceived usefulness and attitude. Similarly, an integrated TAM proposed by Pal and Patra (2021) found a strong correlation between video learning success and student perception [44]. Despite its importance, few studies have used TAM to examine video technology acceptance [40], and research on interactive video specifically is even scarcer. A Scopus search (on February 5, 2023) for "interactive video" and "TAM" returned only one study. Based on this research gap, we propose the following hypothesis:

Hypothesis 1. *The level of students' technology acceptance differs significantly between linear and interactive videos.*

1.1.2. Learning retention

Learning retention is a critical measure of educational performance in both schools and institutions [45]. It is especially important in online education, where course dropout rates are higher than in traditional settings [46]. Retention is defined as the ability to understand and recall new concepts over a period of time, and to apply them in practice [30, 47]. Valderama and Oligo [45] describe it as a process that begins with understanding, moves through processing, and ends with storage in short- or long-term memory [48].

Common indicators for measuring learning retention include the number of students who complete a course [46] and the length of time information is retained [45]. Prior studies suggest that factors like a positive learning experience, engagement, and interaction can improve retention and encourage persistence in online learning [46,49]. On the other hand, ineffective instructional strategies can reduce retention [47], limiting a learner's ability to apply knowledge. To determine if a difference exists between learning retention in linear and interactive videos, we posit the following:

Hypothesis 2. *The level of learning retention differs significantly between linear and interactive videos.*

1.2. Research question development

Recent studies on interactive videos have yielded inconsistent findings regarding user acceptance and learning experience. For example, Rosendahl, Müller, and Wagner [50] found no significant difference in user acceptance between 360° interactive videos and linear videos. Conversely, Haerawan, Cale, and Barroso [51] reported that interactive elements can significantly enhance the online learning experience. These conflicting results highlight the need to explore the underlying mechanisms that influence technology acceptance and learning retention.

On the other hand, Constructivist theory posits that learners actively construct knowledge through interaction and engagement, suggesting that interactive videos may enhance learning by promoting active meaning-making rather than passive reception. From a cognitive theory perspective, particularly the Cognitive Theory of Multimedia Learning, meaningful learning occurs when learners actively process information through selecting, organizing, and integrating content - processes that interactive videos are more likely to stimulate compared to linear formats [52,53]. Finally, TAM explains how perceived usefulness and perceived ease of use influence technology acceptance, providing a theoretical framework to examine how interactive features may shape students' acceptance of educational video technologies [54,55]. Integrating these perspectives clarifies both the learning mechanisms and the acceptance mechanisms underlying the effectiveness of interactive versus linear instructional videos. Therefore, this study also addresses the following exploratory research question:

Research Question: *What mechanisms of interactive video affect students' technology acceptance and learning retention?*

2. Research design and methodology

This study employed a sequential mixed-methods research design, integrating both quantitative and qualitative approaches, as shown in Fig. 2. The quantitative phase, guided by a positivist epistemology, was designed to test the study's two hypotheses. A within-subjects experimental design was used to compare learning outcomes between linear and interactive videos. The qualitative phase, informed by an interpretivist epistemology, aimed to identify the underlying mechanisms of the videos' effects on learners. For this, a phenomenological method was used to understand participants' lived experiences with the videos. This mixed-methods approach allows for a more complete understanding of the phenomena by not only testing for differences (quantitative) but also exploring the reasons behind them (qualitative).

2.1. Quantitative method

To test Hypotheses 1 and 2, a within-subjects experimental design was implemented. Participants were exposed to both the linear and interactive video formats, serving as their own control. This design minimises variability and increases statistical power. Data for technology acceptance and learning retention were collected via questionnaires and multiple-choice tests, respectively. A paired *t*-test was planned for data analysis to compare the mean scores between the two conditions. As shown in Fig. 3, the CONSORT 2010 Flow Diagram illustrates the participant flow throughout the study.

2.1.1. Intervention material description

This study developed the training course Increasing Individual Performance including two part of: an interactive format and a linear (non-interactive) format. Both versions consisted of 14 identical video segments with a total duration of approximately 30 min. Each segment ranged from 2 to 7 min in length in order to optimize cognitive load and maintain learner attention. The instructional content, visual design, narration, production quality, and presenter were held constant across conditions to ensure internal validity. Thus, the only systematic difference between the two conditions was the presence or absence of interactive elements.

As shown Table 1, the interactive version integrated the 14 video segments into a unified digital learning environment enriched with multiple learner-centered interactive features enabling learners to access supplementary explanations, practical examples, or extended content without disrupting the main instructional flow. Formative quizzes, including multiple-choice, were incorporated throughout the video. These assessments provided immediate feedback explaining the correctness of responses, thereby reinforcing conceptual understanding and supporting learning retention. In addition, branching scenarios were implemented at key decision points, allowing learners to select alternative courses of action within simulated workplace situations. Each choice led to consequence-based explanations designed to promote reflective thinking and applied learning. The interactive environment also permitted limited non-linear navigation, enabling learners to customize the order of certain thematic sections according to their preferences or prior knowledge. A visual progress indicator was included to support self-regulated learning and enhance engagement.

In contrast, the linear version presented the same 14 segments sequentially as standalone video clips without any interactive components. Participants in this condition could watch the videos in a fixed order but did not have access to branching pathways, embedded quizzes, clickable supplementary content, navigation flexibility, or progress tracking tools. Therefore, the experimental manipulation specifically isolated the effect of interactivity while controlling for content, duration, segmentation structure, and presenter characteristics.

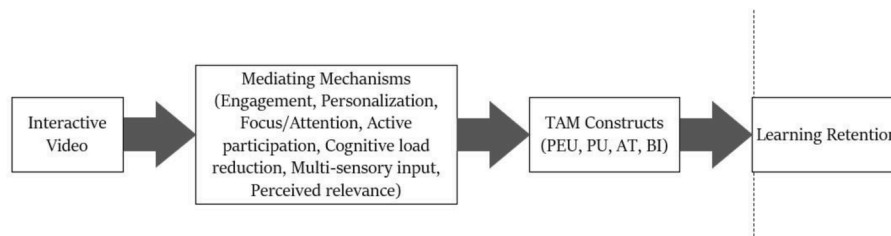


Fig. 2. The research conceptual model.

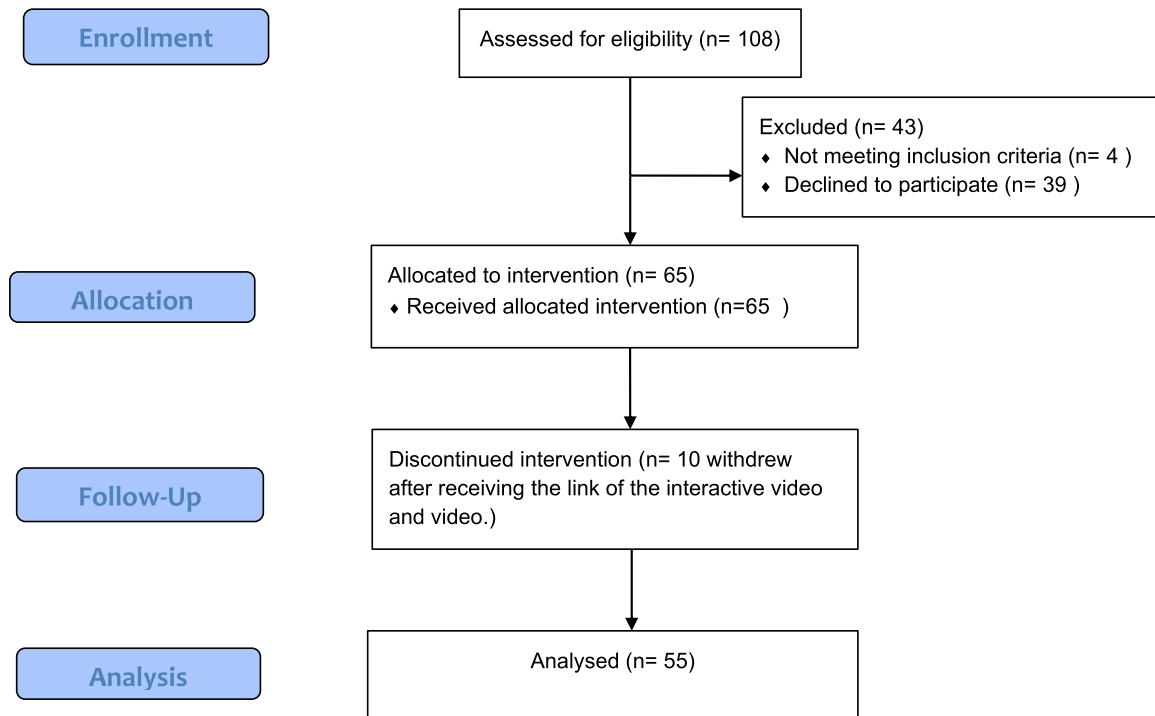


Fig. 3. The CONSORT 2010 flow diagram.

Table 1
comparing the specific features of linear vs interactive videos.

Features	Interactive video	video
Branching pathways	✓	×
Navigation flexibility (the order of sections for watching)	✓	×
clickable supplementary content	✓	×
embedded quizzes	✓	×
progress tracking tools	✓	×
immediate feedback	✓	×
14 video segments	✓	✓
The same presenter	✓	✓
Total Duration	30 m	30 m

2.2. Qualitative method

To address the exploratory Research Question, a phenomenological approach was used. This method allowed for an in-depth exploration of participants' lived experiences and perceptions regarding the interactive video's mechanisms. The primary data collection method was semi-structured interviews, which provided a flexible framework to probe participants' views on how the video's interactive features influenced their learning. A semi-structured interview protocol was developed to explore students' perceptions of linear and interactive instructional videos, focusing on their learning experiences. The interview questions

were designed to uncover detailed insights into their thought processes, feelings, and overall experience, which could not be captured by the quantitative data alone. The protocol also included comparative and reflective questions to identify the underlying mechanisms through which interactive videos may influence both learning outcomes and technology acceptance.

2.2.1. Participants and data collection

The sample size was determined using G*Power version 3.1.9.2, which indicated a required sample of 54 participants based on a medium effect size (Cohen's $d = 0.5$), a significance level of $\alpha = 0.05$, and a statistical power of 95% ($1 - \beta = 0.95$). as shown in Fig. 4, a total of 55 students were recruited using convenience and snowball sampling.

All participants gave informed consent, and the study was approved by the appropriate ethics committee. Participation was voluntary post-graduate student and interested to topic of course, and students were informed of their right to withdraw at any time. To ensure anonymity and confidentiality, all identifiable information was removed from the data before analysis. Data was stored in password-protected files, and responses were aggregated. The sample consisted of 35 females (63.6%) and 20 males (36.4%), with 80% of participants being Master's or Ph.D. candidates.

Participants were emailed links to the videos and questionnaires. To mitigate order effects, the video presentation sequence was randomised. To ensure counterbalancing, we systematically assigned the

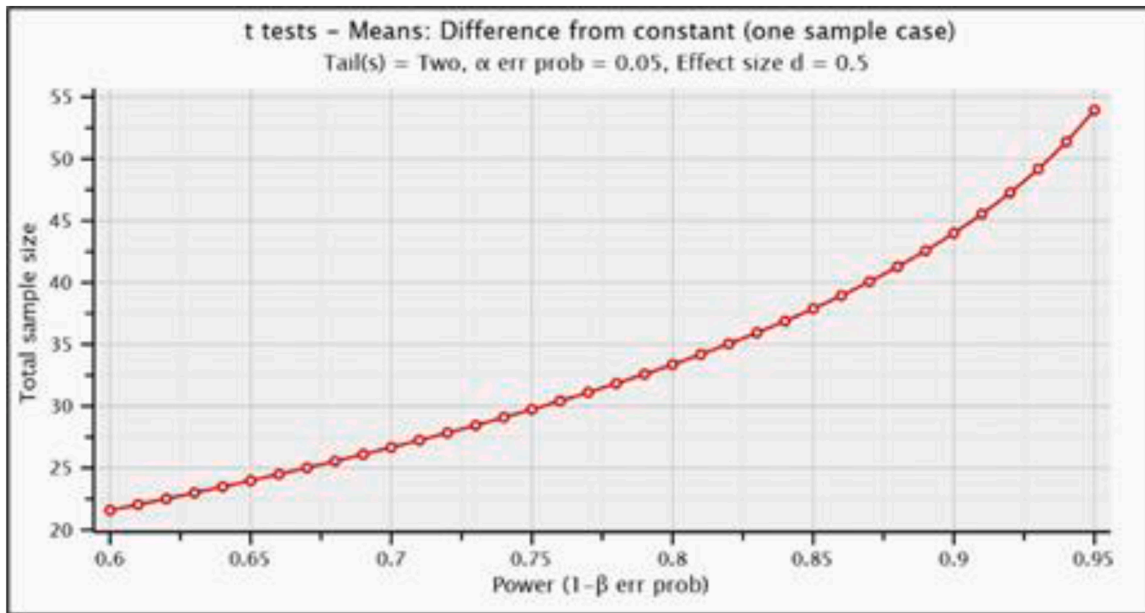


Fig. 4. Calculating the sample size of the study.

intervention order in an alternating sequence: the first participant viewed the linear video followed by the interactive video, the second participant viewed the interactive video followed by the linear video, and this pattern continued for all subsequent participants. After watching each video, participants immediately completed an 8-item multiple-choice test to assess learning retention. Following the tests, they completed the TAM questionnaire to measure technology acceptance for that specific video format.

A subset of twelve participants was then selected for the qualitative phase of the study, where they participated in semi-structured interviews to provide deeper insight into their experience. The interviews were conducted to understand the 'how' and 'why' behind the quantitative results (understanding the mechanisms which affect learners' greater experience with interactive videos). Table 2 lists the qualitative questions based on the TAM model for the semi-structured interviews.

2.3. Data analysis

2.3.1. Quantitative data analysis

A paired *t*-test was used to analyze the quantitative data and determine if there was a statistically significant difference between the learning retention and technology acceptance scores for the linear and interactive videos. This test is appropriate for a within-subjects design because it compares the means of two related groups.

A within-subjects design was employed to control for inter-individual variability in technology acceptance, thereby increasing statistical power and sensitivity to detect differences between video formats. To mitigate potential order effects, the presentation order of the linear and interactive videos was randomised across participants. Approximately half of the participants viewed the linear video first, while the remainder began with the interactive video. This counter-balancing strategy reduced systematic order bias. Furthermore, the duration of exposure was kept consistent across conditions, and no significant order effects were observed in preliminary analyses. The relatively short duration of each condition minimised the likelihood of fatigue or carryover effects.

2.3.2. Qualitative data analysis

The qualitative data from the semi-structured interviews were analyzed using a thematic analysis approach. This process involved transcribing the interviews, coding the data to identify recurring themes

Table 2

TAM qualitative questions for the semi-structured interviews.

Dimension	Item
Perceived Usefulness (PU)	PU1. How much do you perceive you learned with interactive video/linear video? PU2. How actively did you pay attention to the content of the course? PU3. To what extent did each of the content presentation methods make you curious about the subject? PU4. How much did you use additional resources (links and additional explanations)?
Perceived Ease of Use (PEU)	PEU1. In each of the methods, how much freedom did you feel and choose the learning path according to your interests? PEU2. To what extent were you satisfied with the brevity and clarity of each section or topic? PEU3. How satisfied were you with the availability of supplementary resources? PEU4. I find the educational video/interactive video is easy to use.
Attitude (AT)	AT1. How eager were you to see the content of the course to the end? AT2. To what extent do you prefer each of the presentation methods for future courses? AT3. I believe that the educational video/interactive video helps to be more engaged in learning.
Behavioral Intention (BI)	BI1. Are you so tired that you decide to leave the training course? BI2. I intend to use educational video throughout this semester and the next. BI3. I intent repetitively use the educational video as often as possible.

and patterns related to the interactive video's features, and synthesising these themes to address the research question. To ensure inter-rater reliability and transparency in this process, two researchers independently coded the transcripts. The initial open coding phase yielded a total of 42 initial codes. The researchers then compared their independent codebooks to identify discrepancies. Consensus on conflicting codes was reached through iterative discussion and collaborative definition refinement until 100% agreement on the final coding scheme was achieved. The aim was to identify and describe the core mechanisms and experiences that influenced participants' acceptance and retention.

The questions were open-ended to allow participants to elaborate on

their perceptions, while follow-up probing questions were asked when necessary to clarify responses. The core interview questions were as follows: 1) Between interactive video and linear video, which would you prefer to use in the future, and why?, 2) What emotions did you experience while using the interactive video?, 3) How did the interactive video affect your learning process?, 4) How did the interactive video influence the amount or depth of your learning?, 5) What features of the interactive video attracted your attention?

2.4. Trustworthiness

To ensure the rigor and credibility of the qualitative findings, several measures were taken to establish trustworthiness. Credibility was enhanced through member checking, where a summary of the findings was shared with a subset of participants to ensure the interpretations accurately reflected their experiences. Transferability was addressed by providing a thick description of the research context and participant demographics. Dependability was maintained through a detailed audit trail of the research process, and confirmability was established by keeping a reflexive journal to document the researchers' biases and assumptions throughout the study.

3. Results

This section presents the findings of the study, structured to address each hypothesis and the research question. The quantitative results are presented first, followed by the qualitative findings that provide deeper insight into the observed effects.

3.1. Preliminary analysis

First, the internal consistency of the Technology Acceptance Model (TAM) questionnaire was assessed using both Cronbach's alpha and McDonald's Omega. The results showed a high level of internal consistency for both the linear video ($\alpha = 0.898$, $\omega = 0.904$) and interactive video ($\alpha = 0.89$, $\omega = 0.897$) questionnaires. Scores greater than 0.8 are considered to indicate good internal consistency.

To ensure the validity of the paired-samples *t*-test, the normality of the difference scores ($\text{diff_TAM} = \text{TAM_L} - \text{TAM_I}$) was examined using skewness and kurtosis statistics. The difference scores had a mean of -1.842 ($SD = 1.269$), with skewness $= -0.053$ and kurtosis $= -0.306$. As these values fall well within the acceptable thresholds (skewness between -1 and 1 ; kurtosis between -2 and 2), the assumption of normality was satisfied. Bootstrapping procedures were additionally applied to obtain robust confidence intervals, confirming the stability of the mean difference.

Effect size was calculated using Cohen's *d* for dependent samples, defined as the mean of the difference scores divided by their standard deviation ($d = M_{\text{diff}} / SD_{\text{diff}}$). Based on this calculation ($-1.842 / 1.269$), the effect size was $d = 1.45$, indicating a large practical effect in favor of the interactive video condition.

Hypothesis 1. Technology Acceptance.

The descriptive statistics for technology acceptance are summarised in Table 3. The mean acceptance score for the linear video was 3.06 (SD

$= 0.698$), while the mean for the interactive video was significantly higher at 4.24 ($SD = 0.495$).

A paired *t*-test was conducted to compare the means. The results, as shown in Table 4, revealed a significant difference in technology acceptance between the two video formats, $t(54) = -10.31$, $p < .001$. The average difference was -1.18 , with a 95% confidence interval of $[-1.41, -0.95]$. Since the *p*-value is < 0.01 , and the confidence interval does not contain zero, Hypothesis 1 is supported. The negative sign of the mean difference indicates that students had a significantly higher level of technology acceptance for the interactive video compared to the linear video.

As detailed in Table 5, all four dimensions of the TAM - perceived ease of use (PEU), perceived usefulness (PU), attitude (AT), and behavioral intention (BI) - were higher for the interactive video. Behavioral intention had the highest mean score (4.327), while perceived usefulness had the lowest (4.168).

The Fig. 5 displays the scores for each dimension of the TAM model for both video types.

Finally, a correlational analysis showed a non-significant, positive correlation between the technology acceptance of the linear and interactive videos, $r(55) = 0.009$, $p = .950$, as shown in Table 6.

The non-significant correlation between acceptance of the linear and interactive video formats suggests that students' evaluations were context-dependent rather than reflecting a stable, generalized disposition toward video-based learning. In other words, acceptance appears to be shaped by specific features of the instructional format rather than individual-level technological predispositions. This finding supports the interpretation that the interactive design itself, rather than participant characteristics, drove the observed differences.

Hypothesis 2. Learning Retention.

Learning retention was assessed using an 8-item multiple-choice test administered immediately after each video condition. Each item was scored dichotomously (1 = correct, 0 = incorrect), resulting in a maximum possible score of 8 per condition. The questions were directly based on the instructional content presented in the respective videos and were developed in alignment with the stated learning objectives and key concepts. To support content validity, the items were reviewed by two subject-matter experts for relevance, clarity, and coverage, and minor wording revisions were made prior to data collection. Descriptive statistics for both conditions are presented in Table 7.

A paired-samples *t*-test (see Table) indicated that retention scores were significantly higher in the interactive condition ($M = 5.60$, $SD = 1.84$) compared to the linear condition ($M = 4.56$, $SD = 1.94$), $t(54) = -3.59$, $p < .001$. As shown in Table 8, the mean difference was -1.04 (95% CI $[-1.62, -0.46]$). As shown in Table 9, the effect size, calculated using Cohen's *d* was $d = 0.48$, indicating a moderate practical effect. These findings support Hypothesis 2, suggesting that interactive videos lead to improved learning retention relative to linear videos.

3.2. Research question: interactive videos' mechanisms

The qualitative analysis of the semi-structured interviews identified seven key mechanisms that explain why interactive videos positively impacted students' technology acceptance and learning retention. These

Table 3
Paired sample descriptive statistics for linear vs. interactive video acceptance.

		Statistics	Bias	Std. Error	BCa 95% CI Lower	BCa 95% CI Upper
TAM (Linear video)	Mean	3.0629	-0.0023	0.095	2.8886	3.2442
	N	55				
	Std. Deviation	0.69841	-0.00766	0.04129	0.62871	0.7521
TAM (Interactive video)	Mean	4.2436	0	0.0675	4.1164	4.3737
	N	55				
	Std. Deviation	0.49504	-0.00478	0.04146	0.41612	0.56331

Table 4

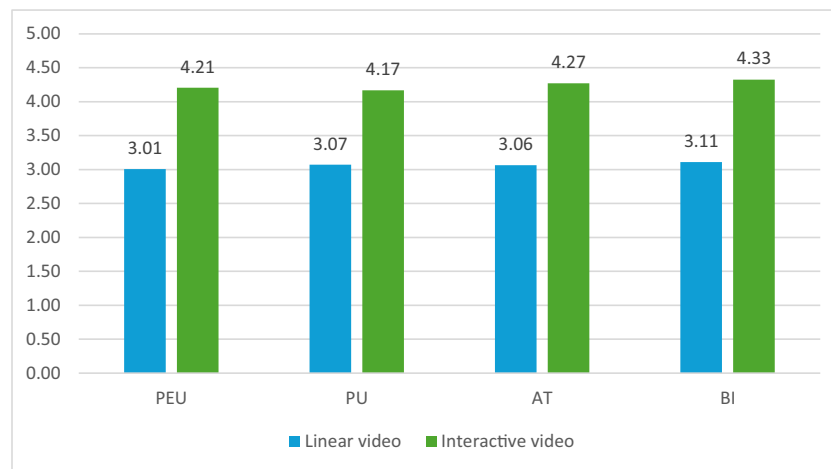
Paired samples test between linear and interactive videos.

Column 1	Mean	Std. Deviation	Std. Error Mean	BCa 95% CI Lower	BCa 95% CI Upper	t	df	Sig. (2-tailed)
Difference	-1.181	0.85	0.11	-1.41	-0.95	-10.31	54	0

Table 5

Descriptive statistics of TAM elements.

Video Type	TAM Elements	N	Minimum	Maximum	Mean	Std. Deviation
Linear	Perceived Ease of Use (PEU)	55	1.33	4.67	3.0061	0.78171
	Perceived Usefulness (PU)	55	2	4.25	3.0727	0.64491
	Attitude (AT)	55	1.5	4.5	3.0636	0.80539
	Behavioral Intention (BI)	55	1	5	3.1091	0.69841
Interactive	Perceived Ease of Use (PEU)	55	3	5	4.2061	0.51516
	Perceived Usefulness (PU)	55	3	5	4.1682	0.52054
	Attitude (AT)	55	3	5	4.2727	0.57589
	Behavioral Intention (BI)	55	2	5	4.3273	0.79476

**Fig. 5.** TAM scores by dimension and video type.**Table 6**

Correlations between linear and interactive videos.

N	Correlation	Sig.	Bias	Std. Error	BCa 95% CI Lower	BCa 95% CI Upper
55	0.009	0.95	-0.003	0.143	-0.288	0.286

mechanisms – as shown in Table 10 - highlight how interactive elements facilitate a more engaging and effective learning experience.

Mechanism 1. Perceived Usefulness Enhancement.

Interactive videos systematically increase Perceived Usefulness by demonstrating the immediate real-world relevance of the learning material. Instead of viewing the video as a static chore, students recognise it as a practical utility. As one participant noted, the inclusion of “*hyperlinks and external resources showing how each concept applied in real situations... made the online session feel more useful than just reading slides*”. Crucially, this qualitative insight explains the high quantitative TAM scores observed in Phase 1, proving that when educational technology contextualises theory, students attach a higher functional value to the

Table 7

Descriptive statistics and bootstrap confidence intervals for learning retention scores by video condition.

Video Condition	Statistic	Value	Bootstrap Bias	Standard Error	BCa 95% CI Lower	BCa 95% CI Upper
Learning Retention (Linear)	Mean	4.56	.01	.26	4.04	5.07
	N	55	—	—	—	—
	Std. Deviation	1.941	-0.024	.129	1.649	2.176
Learning Retention (Interactive)	Mean	5.60	.00	.24	5.11	6.09
	N	55	—	—	—	—
	Std. Deviation	1.842	-0.025	.147	1.526	2.106

Table 8

Paired-samples t-test results for learning retention differences between linear and interactive video conditions.

	Mean	Std. Deviation	Std. Error Mean	BCa 95% CI Lower	BCa 95% CI Upper	t	df	Sig. (2-tailed)
Difference	-1.036	2.143	.289	-1.616	-0.457	-3.587	54	.001

Table 9

Effect size (cohen's d) with bootstrap confidence intervals for the condition difference.

	Standardizer	Point Estimate	BCa 95% CI Lower	BCa 95% CI Upper
Cohen's d	2.143	−0.484	−0.761	−0.202

Table 10

Seven mechanisms of interactive videos influencing technology acceptance and learning retention.

	Mechanism	Description	Key Interactive Triggers
1	Perceived Usefulness Enhancement	Aligns abstract concepts with real-world utility, increasing perceived value.	Hyperlinks, external resource extensions
2	Perceived Ease of Use Optimisation	Minimises technical friction and cognitive tracking effort.	Segmented timelines, click-to-jump navigation
3	Positive Attitude Cultivation	Shifts user perception from passive consumption to conversational engagement.	Interactive dialogue prompts, choice architecture
4	Active Engagement & Participation	Enforces active cognitive processing during information delivery.	In-video quizzes, pop-up questions, branching choices
5	Personalised Learning Support	Grants user autonomy over pacing, review, and depth of content.	Playback speed controls, skippable sections, rewind paths
6	Attention & Focus Retention	Interrupts passive viewing intervals to mitigate cognitive drifting.	Fixed-interval interactive prompts, knowledge checks
7	Multi-Sensory Processing	Co-activates visual, auditory, and kinesthetic learning pathways.	Synchronised audiovisual cues, physical click actions

tool.

Mechanism 2. Perceived Ease of Use Optimisation. In addition to utility, the structural design of the media plays a pivotal role in reducing technical friction and enhancing user experience. One participant highlighted this seamless navigation, stating that the system was “easy to follow as I could jump to the parts I needed,” and that the “timing layout of the interactive video made it simple to stay on track without getting lost.” This highlights how explicit user control over chronological navigation alleviates cognitive tracking effort. By removing the traditional frustration of manually scrubbing through long-form timelines, the user interface maximises perceived ease of use, freeing up finite cognitive resources for actual content absorption.

Mechanism 3. Positive Attitude Cultivation.

The combination of high utility and low technical frustration directly influences the learner's emotional disposition toward the subject matter. This interactive approach fundamentally changed how students felt about the material; as one interviewee shared, “I used to find this topic dull, but the interactive video made it come alive... The way it was presented felt more like a real interactive conversation than a lecture.” Within the TAM framework, affective responses are critical precursors to sustained technology adoption. By reframing a passive monologue as a bilateral conversation, the technology successfully cultivates positive attitudes, which effectively bridges the gap between initial utility and long-term behavioural intent to learn.

Mechanism 4. Active Engagement & Participation.

Beyond technology adoption, the system's structural architecture forces a disruption of traditional, passive viewing dynamics. The integration of features requiring regular physical and mental inputs ensured that students remained active participants. A participant reflected on this shift, noting, “I wasn't just watching, I had to make choices and answer

questions, which kept me thinking and actually involved in what I was learning.” This active manipulation transforms the video from an isolated background streaming experience into an intensive cognitive workspace. Forcing immediate retrieval and decision-making ensures that exposure time translates directly into active comprehension and knowledge retention.

Mechanism 5. Personalised Learning Support.

This active engagement is further enhanced by the platform's capacity to support student autonomy and cater to differentiated learning speeds. This flexibility allowed the technology to mimic a bespoke, one-on-one tutorial environment. An interviewee explained, “I liked that I could go back, skip ahead, or dive deeper into parts I found harder - it felt like the video adapted to how I learn best.” From an acceptance perspective, customisation satisfies the core psychological need for autonomy. By transferring pacing control from the instructor to the learner, the system accommodates idiosyncratic learning curves, reinforcing both perceived ease of use and individual learning effectiveness.

Mechanism 6. Attention & Focus Retention.

By maintaining this personalised momentum, the system effectively combats the common digital learning pitfalls of attentional drift and continuous partial attention. Regular interactive checkpoints served as cognitive anchors throughout the session, preventing external distractions. One student noted the stark contrast with conventional delivery methods, stating, “Because the video asked me to interact every few minutes... I stayed focused. I couldn't just zone out like I sometimes do in regular online or face-to-face lectures.” In online learning contexts, “zoning out” is a primary barrier to information retention. By introducing mandatory, time-delimited micro-interactions, the video provides structural interruptions that reset the student's attention span, guaranteeing consistent engagement over extended periods.

Mechanism 7. Multi-Sensory Processing.

Finally, the cognitive benefits of these focus-retaining elements are anchored by the simultaneous co-activation of dual-coding pathways through a multimodal interface. The sensory-rich environment was highly valued for its capacity to reinforce mental models. As a participant concluded, “Seeing the visuals, hearing the explanation, and clicking through the content helped me take it in better... It felt like I was more involved in the learning, not just passively watching.” This reflects the core principles of cognitive load theory, where learning is optimised when visual and auditory channels are accompanied by kinesthetic action. The physical act of clicking, tightly synchronised with audiovisual cues, deepens retention by shifting the student away from a passive spectator role and into an active cognitive producer.

4. Discussion and conclusion

This study makes three key contributions to the field of online learning and educational technology within the context of the post-graduate population studied. First, the findings suggest that interactive videos can significantly enhance students' technology acceptance compared to linear video formats. Second, the results indicate that for this cohort, interactive videos are associated with improved learning retention.

The findings of this study regarding the potential efficacy of interactive videos highlight the importance of interaction in online learning environments for higher education students. This result aligns with research demonstrating the impact of interaction in digital settings on self-directed learning [56,57] and with studies on the effect of interactive whiteboards on self-efficacy and academic achievement [58], further underscoring the significance of interactivity.

However, the most important contribution lies in the identification of seven underlying mechanisms that help explain how these positive outcomes may be achieved. This is a critical finding because it moves

beyond simply observing the benefits of interactive videos and provides practical guidance on how these tools can be designed to be effective. Without incorporating these mechanisms - such as promoting active participation, supporting personalization, and enhancing focus - educators and designers risk deploying interactive videos that fail to deliver meaningful learning gains. By linking these mechanisms to relevant learning theories, the study reinforces the idea that educational technology must be driven by pedagogical principles [59], echoing calls in the literature to treat technology as a medium for instruction rather than an end in itself [60].

4.1. Underlying mechanisms and educational theories

The seven mechanisms identified in this study are deeply rooted in established educational theories, providing a strong theoretical foundation for their influence. As shown in Table 11, a direct link can be drawn between each mechanism and a supporting theory.

The Technology Acceptance Model (TAM) directly supports two key mechanisms: enhancing the perceived usefulness and ease of use of the content. According to TAM, these two perceptions are central to a learner’s willingness to adopt and engage with new technologies [61]. Similarly, fostering positive attitudes aligns with the Affective Domain of Bloom’s Taxonomy, which recognizes the importance of emotional engagement in motivating learners and shaping their disposition toward the content [62].

The mechanism of increasing student engagement and active participation is supported by Constructivist Learning Theory, which emphasizes the importance of learners actively constructing knowledge through interaction. The ability of interactive videos to support personalized learning experiences aligns with the principles of Universal Design for Learning (UDL), which advocates for flexible, learner-centered approaches that accommodate individual needs [63].

Furthermore, Cognitive Load Theory explains how breaking content into smaller, interactive chunks can help sustain focus and attention [64], reducing cognitive overload in online environments. Finally, the mechanism of engaging multiple sensory channels is supported by Mayer’s Multimedia Learning Theory, which posits that learners retain information more effectively when both visual and auditory modalities are integrated [65].

These well-established pedagogies provide a reassuring framework for both students and teachers, confirming that these learning technologies are designed and implemented based on sound principles and guidelines. When designing an interactive video, these theories can guide designers on how to create effective interactions.

4.2. Interdisciplinary approach to educational technology

The findings of this study reinforce the need for an interdisciplinary approach to designing and implementing educational technology. The

participants in this study, many with strong backgrounds in education, were able to identify and articulate solutions that were rooted in educational theory. This stands in stark contrast to technologies designed solely by computer science experts, which often fail to improve learning quality due to a lack of pedagogical expertise [66,67].

As scholars have noted, an effective educational technologist must be an instructional designer [68] and identified learner experience and their perceptions of various types of interaction in online learning environment [69]. Additionally, effective design requires considering not only technological and pedagogical aspects but also psychological dimensions [70] and sociological theories [71]. Solutions for creating effective interactions must be grounded in pedagogy [72]. Specifically, the learner-centered multimedia design principle must be designed in distance education [73]. Furthermore, concepts from cognitive and educational psychology are critical for addressing issues related to attention, memory, motivation, and cognitive load [74]. Therefore, the design and implementation of learning technology is a process that must involve multiple disciplines, including education, psychology, and sociology, with leadership from pedagogical experts. This collaborative approach is essential to ensure that new technologies effectively support, rather than hinder, the learning process.

4.3. Implications

This study demonstrates several methodological and practical strengths. First, the mixed-methods design is a key contribution, as it combines causal inference from a rigorously controlled within-subjects experiment with rich qualitative insights that explain the underlying mechanisms of effectiveness. Second, the a priori power analysis using G*Power ensured an adequately powered sample, strengthening the statistical rigor and credibility of the findings. Third, the within-subjects design increased statistical power while controlling for individual differences. Beyond methodological rigor, the study offers a clear practical contribution through the seven-mechanism framework, which provides actionable design principles for educators and instructional designers. Theoretically, the integration of TAM, cognitive theory, and constructivist perspectives reflects a strong interdisciplinary foundation.

The primary theoretical implication of this research is the identification of seven key mechanisms that explain how interactive videos positively impact students’ technology acceptance and learning retention. These mechanisms serve as critical design principles that must be met for interactive videos to be effective learning tools. By grounding these conditions in established educational theories - such as the Technology Acceptance Model and Constructivist Learning Theory - the study provides a robust theoretical framework for future research and development. This work highlights the importance of an interdisciplinary approach to educational technology design, emphasizing that pedagogical principles must guide technological innovation.

From a practical perspective, the findings offer a prescriptive framework for educators and instructional designers. To operationalize the seven-mechanism framework, designers should adhere to the following specific guidelines:

- **Interactivity Frequency:** To maintain focus and prevent “zoning out” (Mechanism 6), videos should include at least two interactive decision points or knowledge checks every 5 to 10 min.
- **Segmenting and Navigation:** Content should be segmented into digestible chunks of 6 to 10 min, with a clear, clickable navigation menu (Mechanism 2) that allows learners to skip or revisit sections based on their prior knowledge.
- **Contextual Grounding:** To enhance perceived usefulness (Mechanism 1), designers should incorporate at least one external hyperlink or real-world case study resource per video module to ground theoretical concepts in practice.
- **Multisensory Engagement:** Every module should require at least one tactile interaction (e.g., a drag-and-drop activity or a branching

Table 11
Mapping mechanisms and educational theories.

Mechanism	Supporting Educational Theory
Enhancing students’ perception of the usefulness of the learning content	Technology Acceptance Model (TAM)
Improving their perception of the content’s ease of use	
Fostering more positive attitudes towards the subject matter	Affective Domain of Bloom’s Taxonomy
Increasing student engagement and active participation	Constructivist Learning Theory
Supporting personalized learning experiences	Universal Design for Learning (UDL)
Improving focus and attention during learning	Cognitive Load Theory
Engaging multiple sensory channels in the learning process	Multimedia Learning Theory (Mayer)

choice) in addition to auditory and visual input to ensure the student is an active participant rather than a passive viewer (Mechanism 7).

These videos can significantly enhance flipped learning experiences, where students engage with content before class to free up class time for discussion and questions. By using interactive videos as the pre-class learning content, educators can increase students' technology acceptance and learning retention, leading to more successful and engaging flipped classrooms. Additionally, these findings provide quantifiable standards for the expanding interactive video industry, offering a set of standards for developing more effective and pedagogically sound educational products [59,60].

4.4. Limitations and future research

This study has several limitations that should be considered when interpreting the findings. First, the video content was focused on a single domain (personal development) within an informal learning context. While there is no evidence to suggest this choice impacts effectiveness, the results may not be fully transferable to formal academic settings or diverse disciplines such as STEM or the humanities. Second, the study was conducted at a single university using convenience sampling with a gender imbalance (63.6% female) and a focus on postgraduate students, which may introduce selection bias and limit generalizability to other demographic compositions.

Third, learning retention was assessed immediately following the intervention; therefore, the results do not account for long-term retention or the sustained application of knowledge over time. Finally, the qualitative findings may have been influenced by potential social desirability bias during interviews. Furthermore, the absence of objective behavioral data - such as click-stream patterns, interaction frequency, or precise time-on-task metrics - limits the ability to triangulate self-reported perceptions with observable engagement metrics.

4.5. Future research

Building on these findings, future research should seek to validate the seven-mechanism framework across broader contexts and more diverse student populations to enhance generalizability. Longitudinal

studies are particularly needed to determine if the observed gains in technology acceptance and learning retention persist over time. To address the limitations of self-reported data, future investigations should incorporate learning analytics and objective behavioral metrics to provide a more granular understanding of how students interact with specific video features.

Furthermore, researchers should explore how specific interactive elements can be purposefully designed to align with core educational theories. Key questions remain regarding how these interactions can be optimized for maximum cognitive efficiency and how educators and instructional designers can best collaborate to integrate pedagogical and technological expertise. Finally, future work should examine the role of interactive video in motivating learners within particularly challenging or abstract subject matters, moving beyond "what" works to deeply investigate the "how" of impactful digital learning.

CRediT authorship contribution statement

Hamed Hosseini Zarrabi: Writing – review & editing, Writing – original draft, Visualization, Formal analysis, Data curation, Conceptualization. **Morteza Rezaei-Zadeh:** Writing – review & editing, Validation, Supervision, Methodology, Formal analysis, Conceptualization. **Reza Nabovati:** Software, Methodology, Formal analysis.

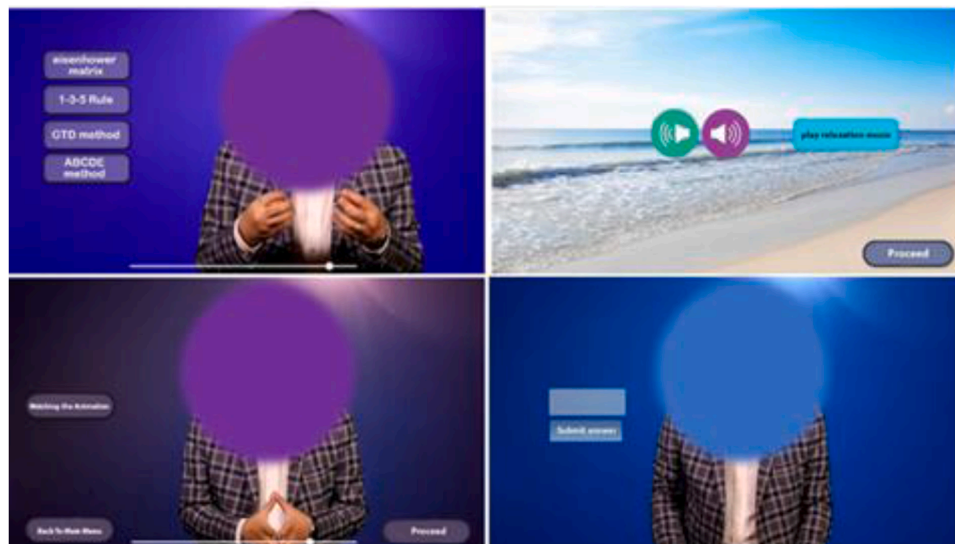
Declaration of competing interest

Authors declare that they have neither any competing interests nor any affiliations with or involvement in any organization or entity with any financial interest (such as honoraria; educational grants; participation in speakers' bureaus; membership, employment, consultancies, stock ownership, or other equity interest; and expert testimony or patent-licensing arrangements) in the subject matter or materials discussed in this manuscript.

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Appendix A. some screenshots from the interactive video



Appendix B. Demographic of participants

Demographic Information	Frequency	Percent
Gender		
Female	35	63.60%
Male	20	36.40%
Age		
19–27	23	41.80%
27–35	18	32.70%
35–43	13	23.60%
43–52	1	1.80%
Education		
Undergraduate	11	20%
Master's students	22	40%
PhD. candidates	22	40%

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